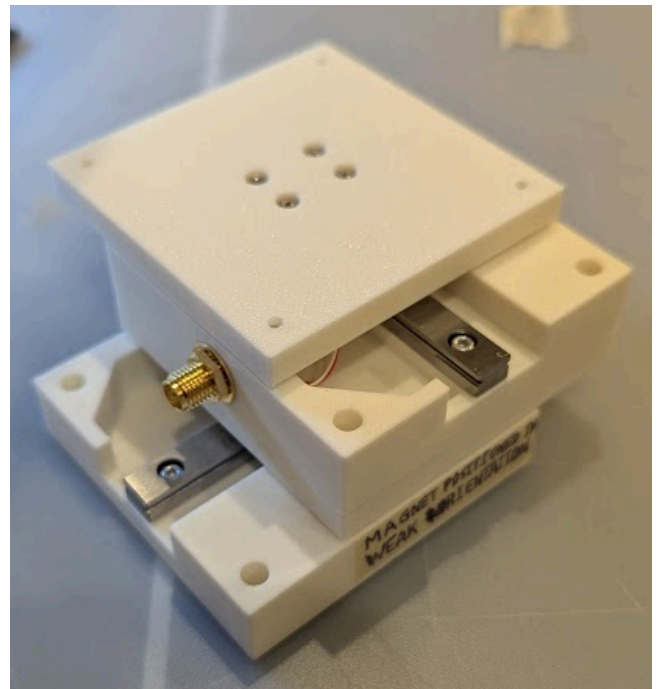
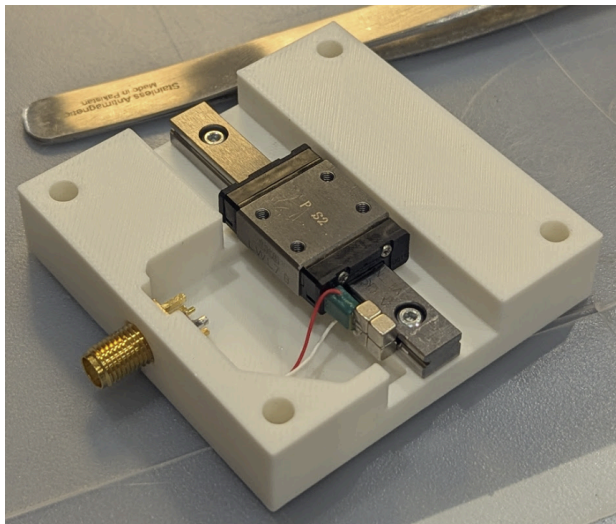
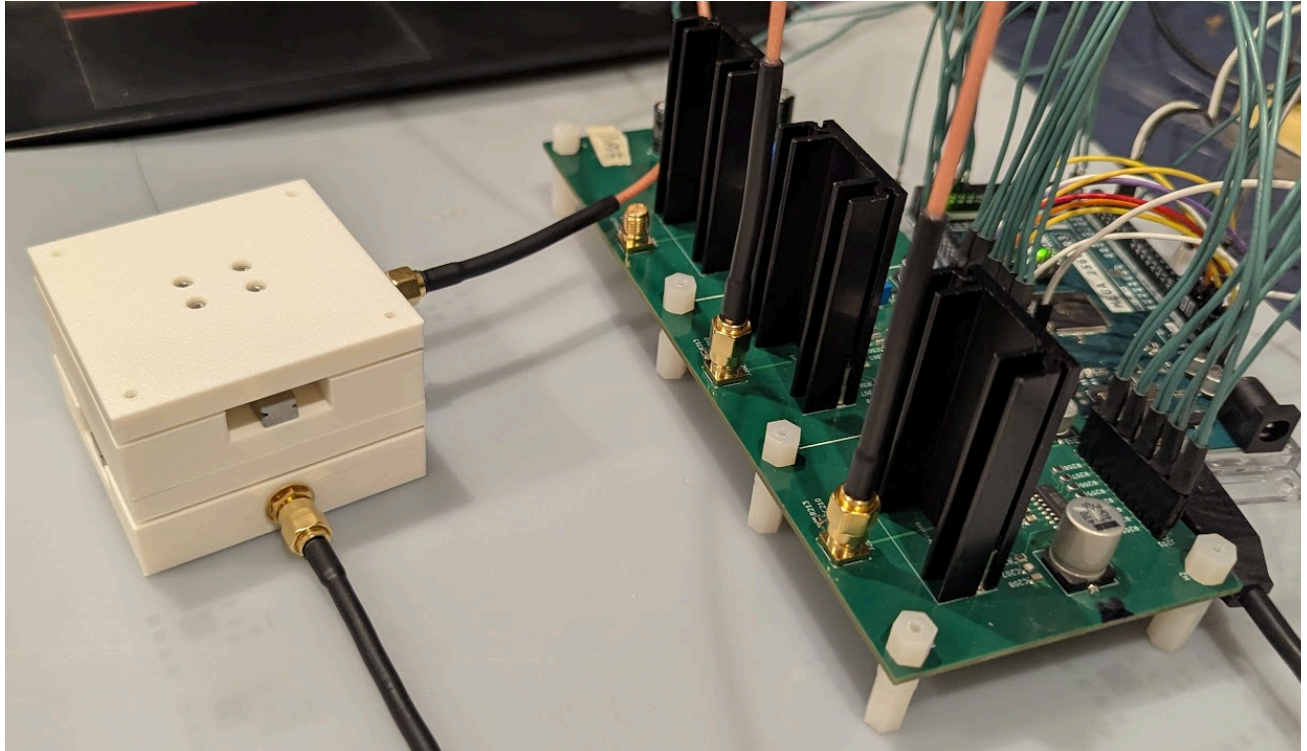
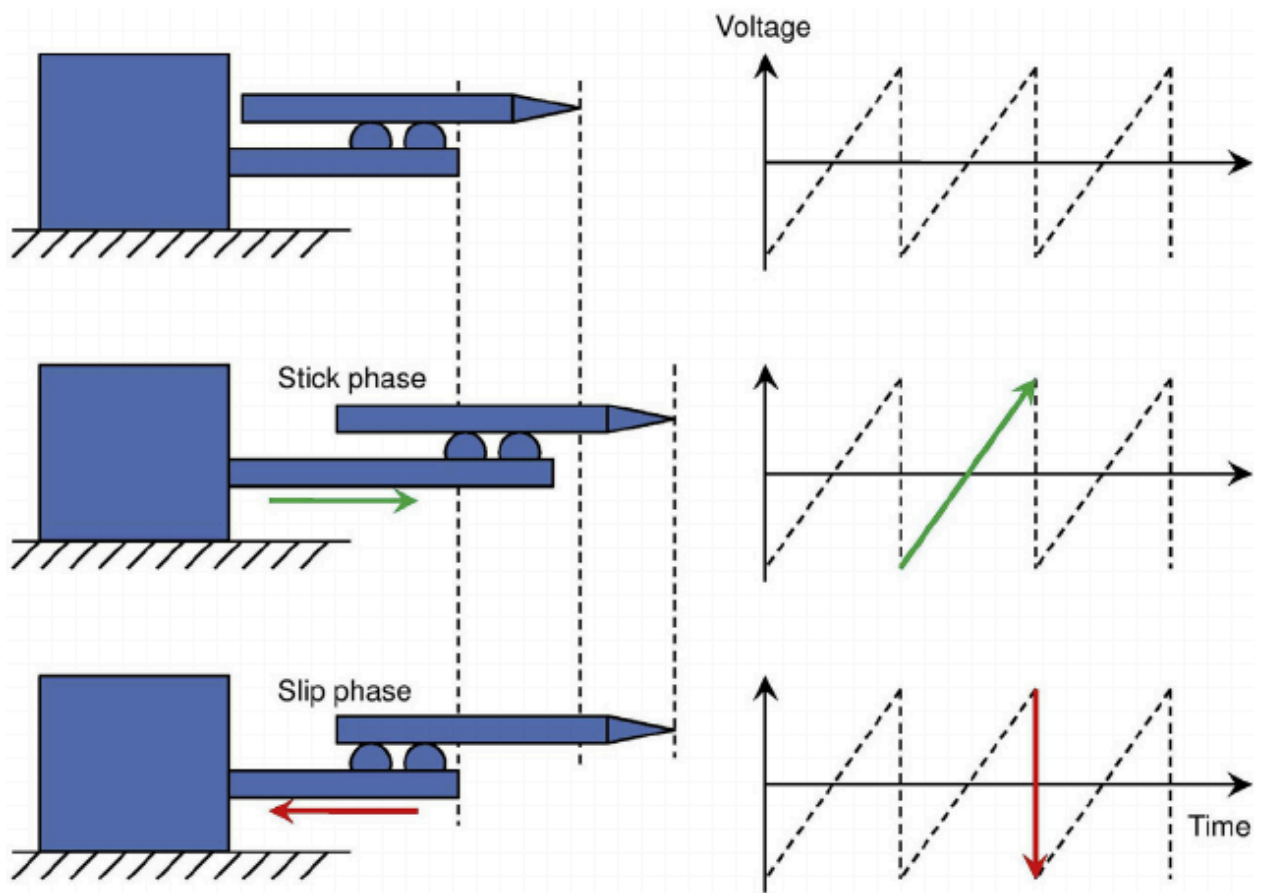


Radu Dehel - Portfolio

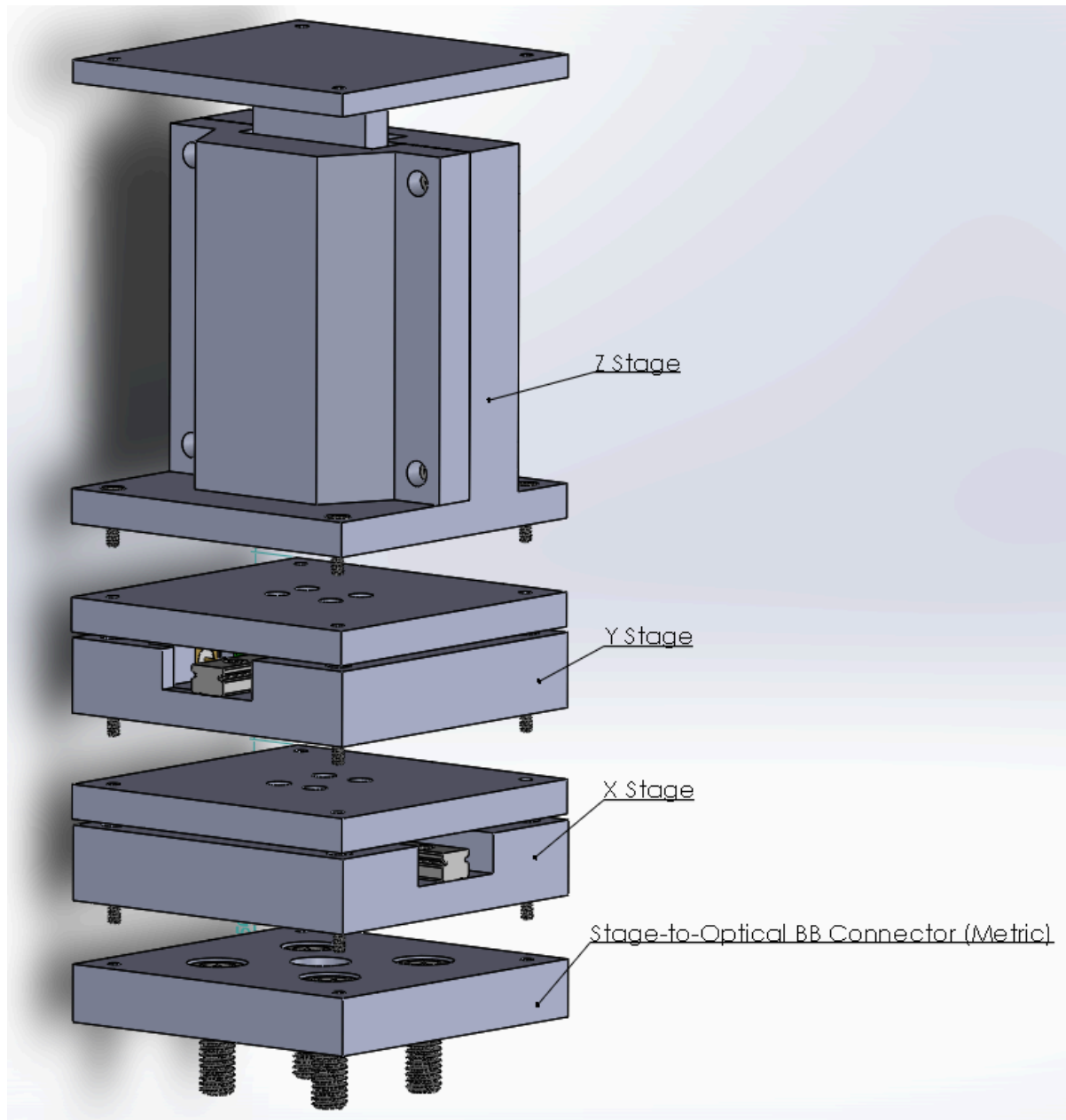
Piezo-nanopositioner wafer stage (Hacker Fab team)



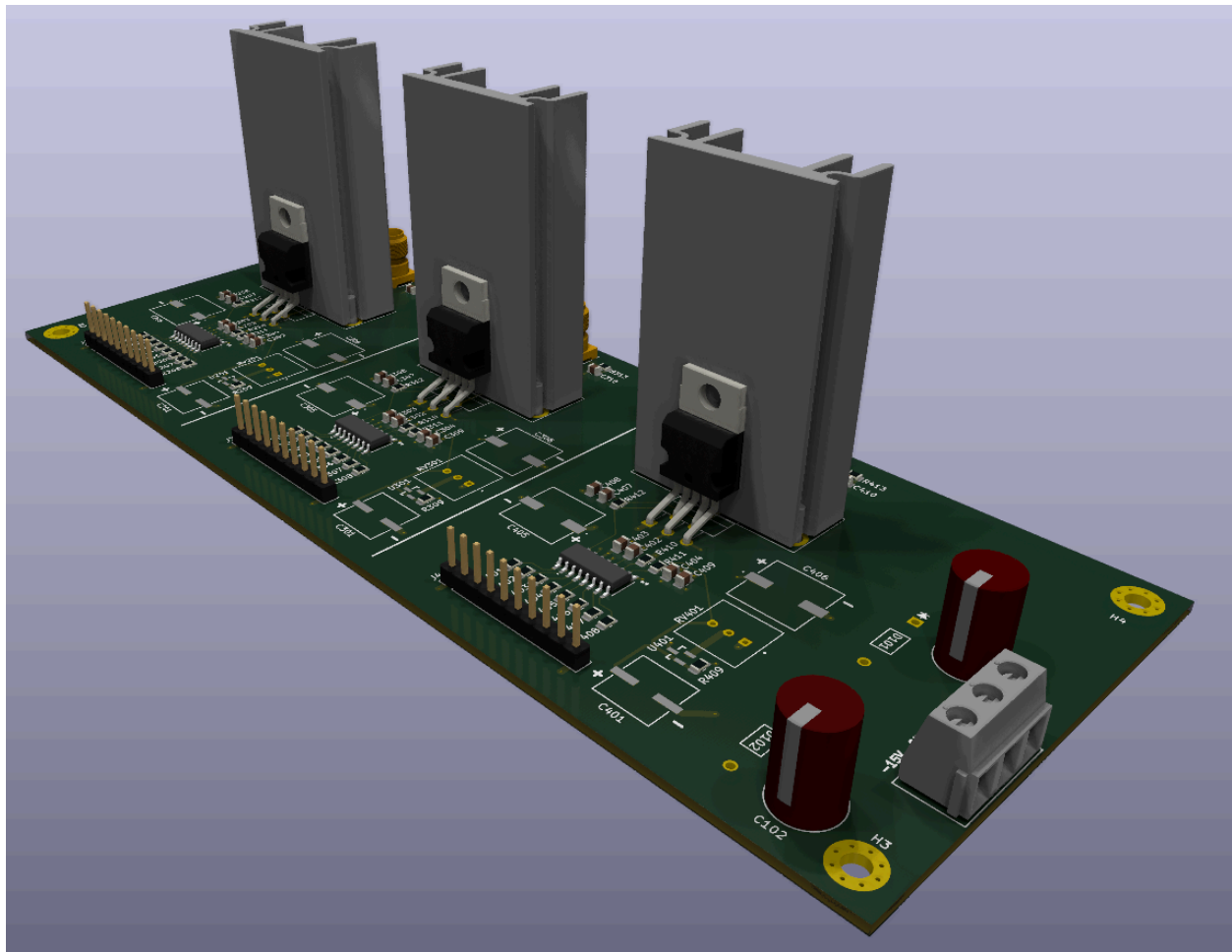
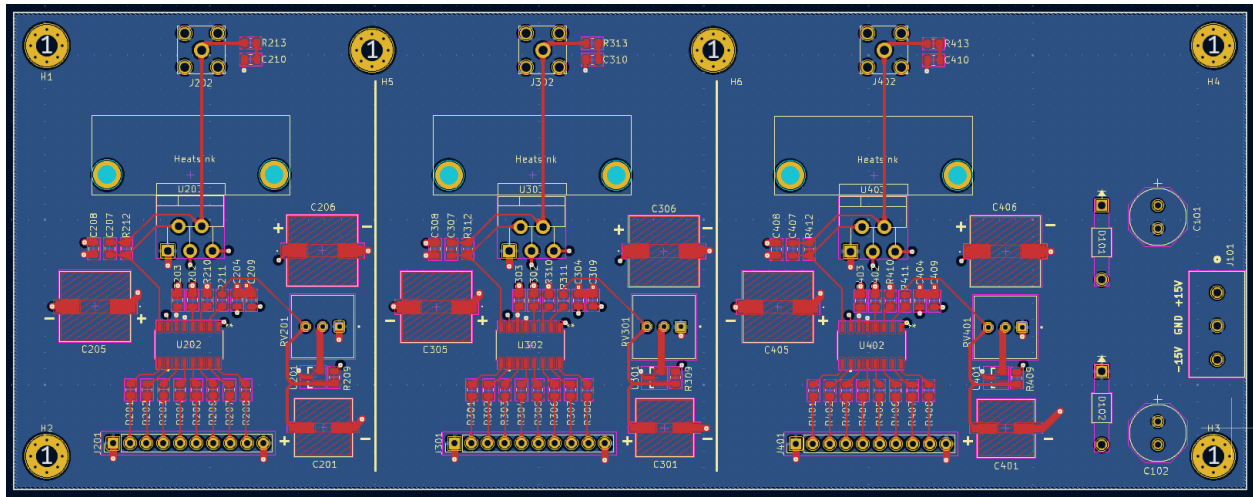
Stick slip mechanism explained:



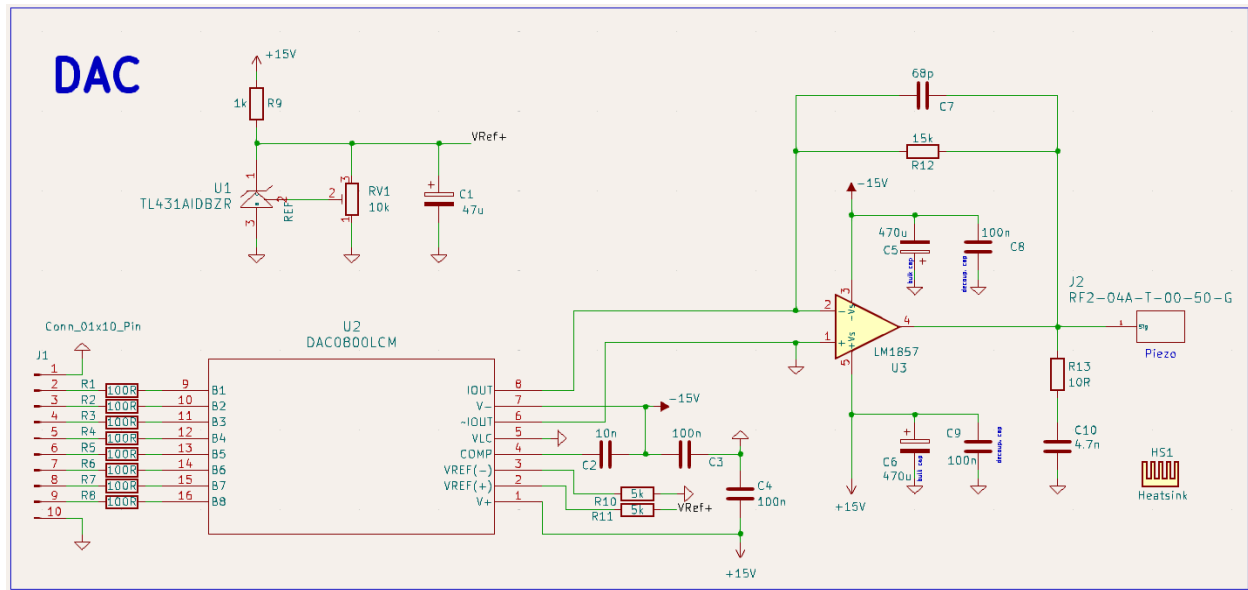
CAD (SolidWorks):



PCB Design (KiCad):



Schematic Design (KiCad):

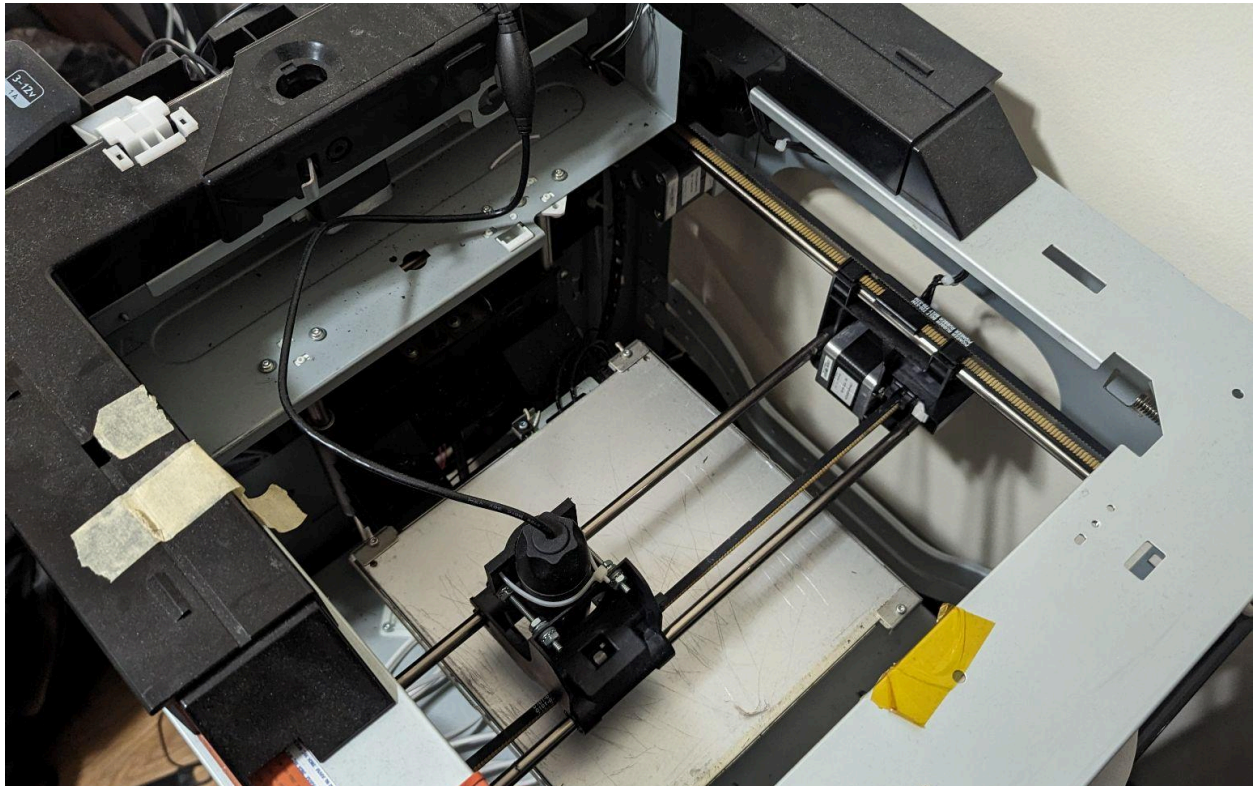


Github: <https://github.com/Waterloo-Hacker-Fab/Hardware/tree/master/Piezo-Nanopositioner-V1>

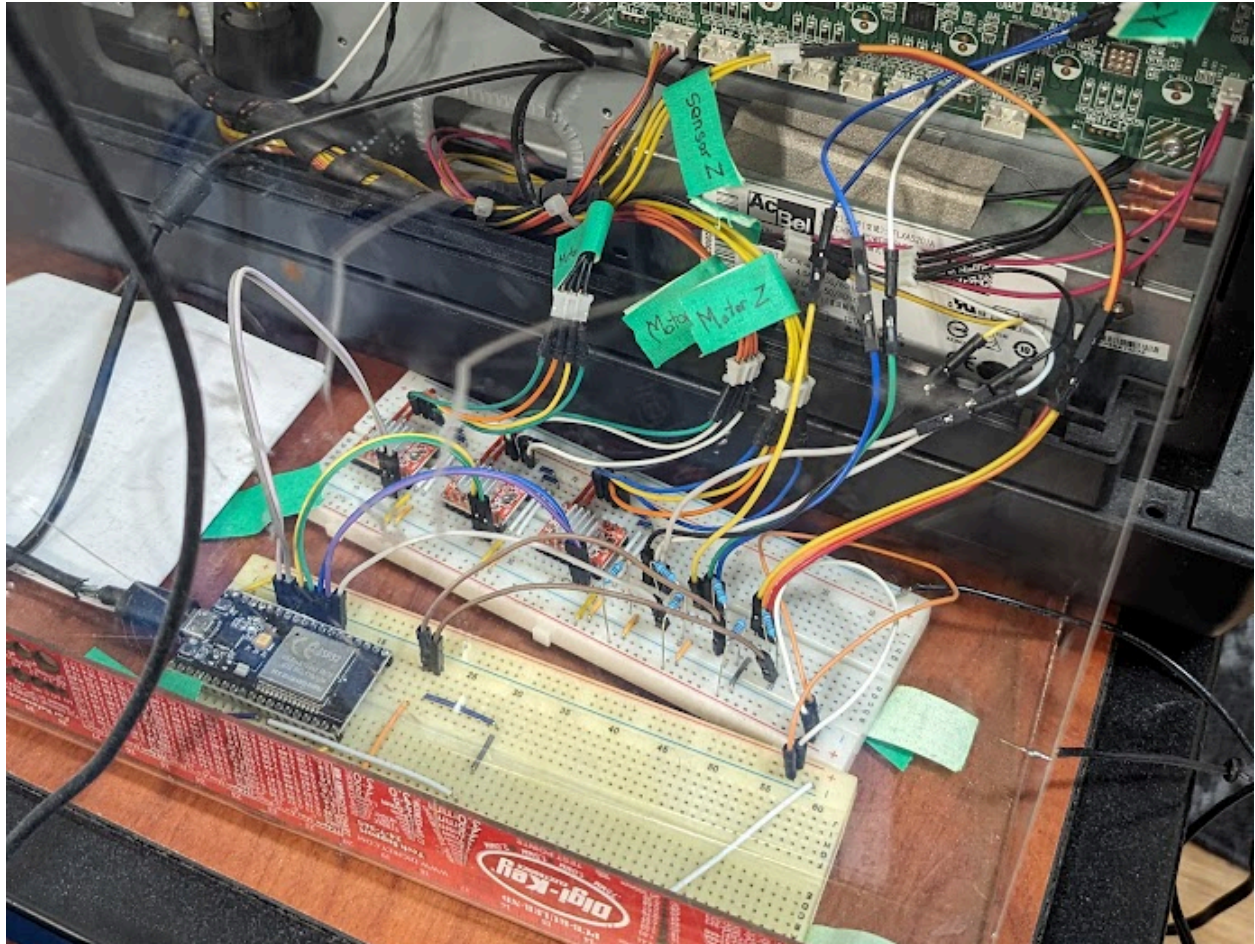
PCB Defect Detection Robot

- **Aim:** Make an open source platform for creators/tinkers at home who want to produce and visually inspect their PCB's for defects at a small-scale manufacturing level.
- Attempted to solve this problem by using a combination of cheap hardware which can easily attach/dismount to a 3D printer.
- NOTE: Defect detection and computer vision was the main focus of this project, the hardware is not amazing and may not be too exciting.

Picture of the head of the 3D printer where the camera now replaces the nozzle

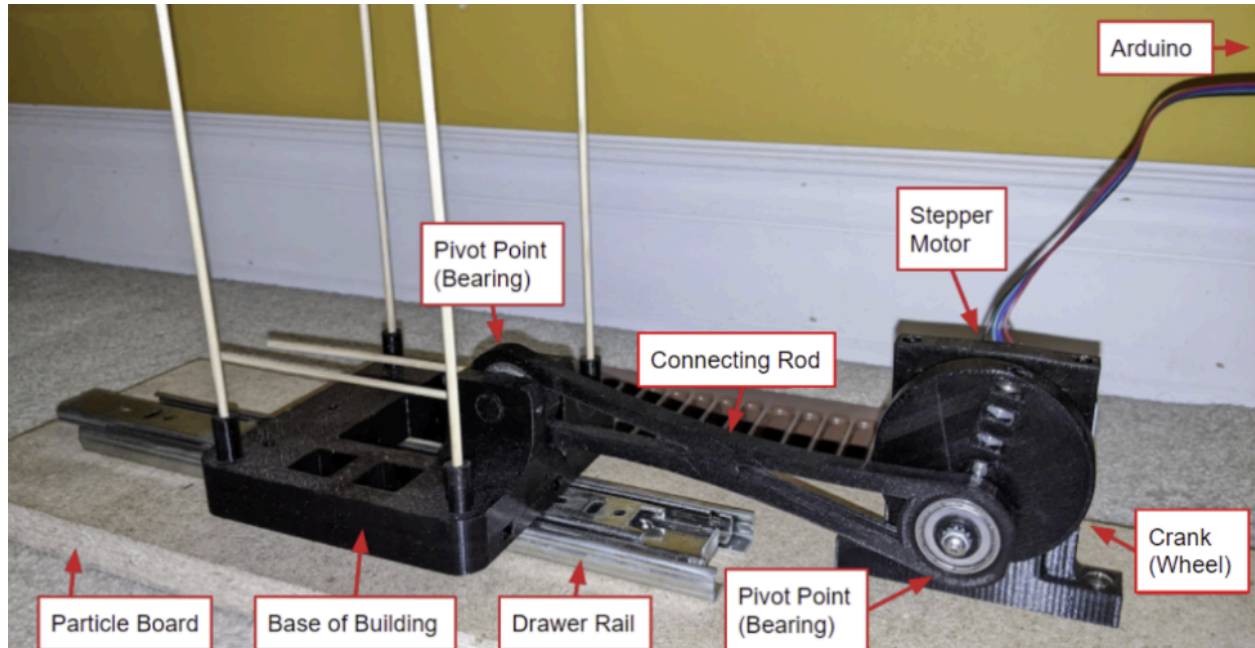


-Current setup of controlling 3D printer head in X,Y,Z directions remotely, using an ESP32 communicating to a computer wirelessly using MQTT protocol.



Tuned Mass Damper system experiment:

- Aim: Analyze the effectiveness of pendulums for damping of vibrations in buildings (etc: earthquakes).
- Made a test rig that vibrates building with pendulum on top, accelerometer at top collects vibration data.
- Arduino/C++ for data processing/automation/motor control
- Performed automated test fixture design to improve data collection speeds



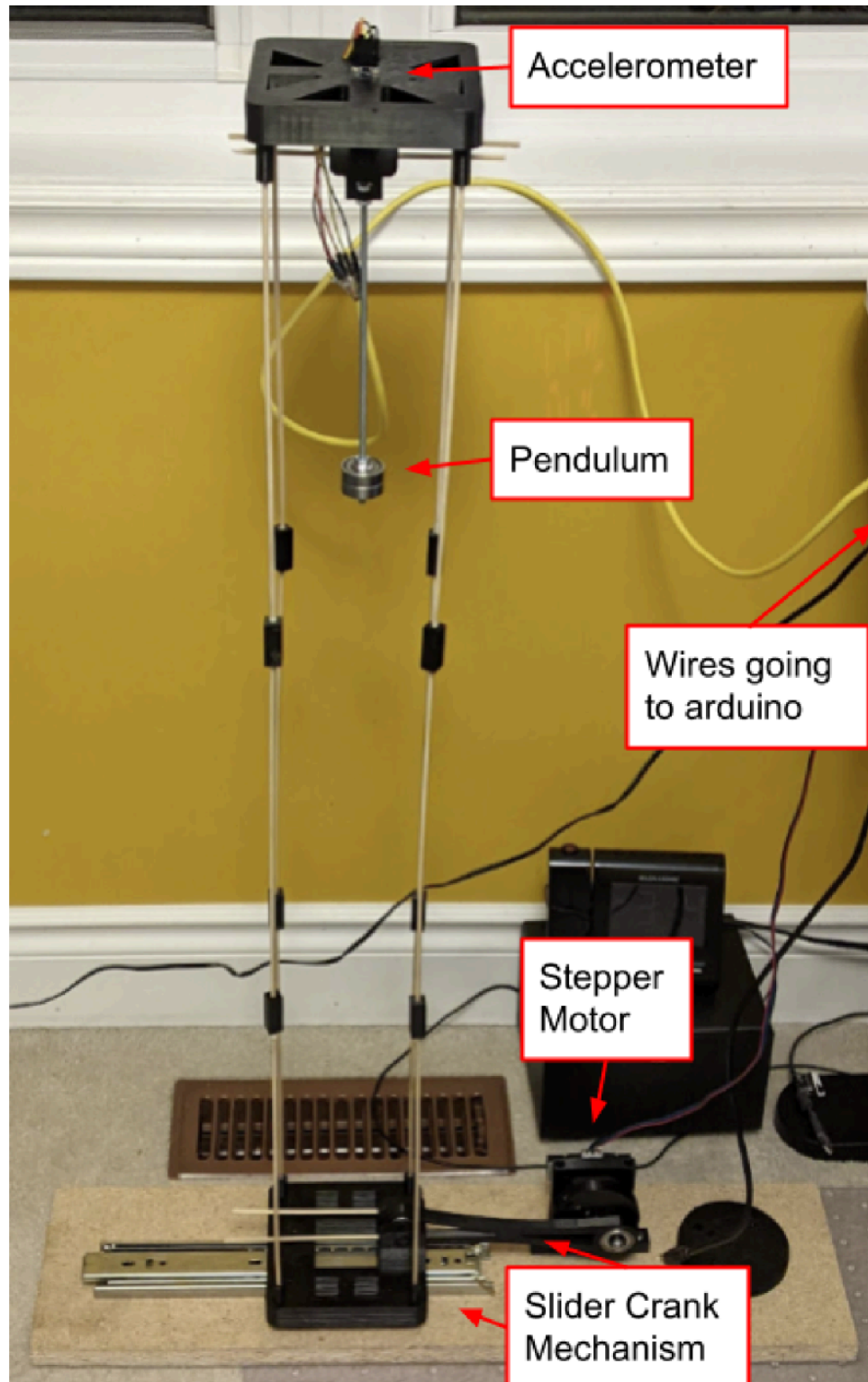


Figure 12: *Picture of final testing setup*

Electric Generator Experiment:

- Made a test rig to conduct an experimental study on the improvements of a generator.
- Used Solidworks for CAD and my own personal 3D printer.
- Arduino/C++ for data processing/automation/motor control.
- Performed automated test fixture design to improve data collection speeds):

